JURONG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATIONS
Higher 2

CANDIDATE NAME

CLASS

BIOLOGY 9744/02

Paper 2 Structured Questions **25 August 2017**

Candidates answer on the Question Paper. **2 hours**

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your name and class in the spaces at the top of this page.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
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This document consists of **25** printed pages and **1** blank page.

[Turn over

Answer **all** questions.

1 The plasma membrane Ca^{2+} ATPase (PMCA) is a vital transport protein that regulates the amount of Ca^{2+} within eukaryotic cells. In humans, there is a very large transmembrane electrochemical gradient of Ca^{2+} driving the entry of Ca^{2+} into cells, yet low intracellular concentrations of Ca^{2+} are maintained by PMCA.

Fig. 1.1 shows two conformations that PMCA interconverts between, depending on whether Ca^{2+} is bound.

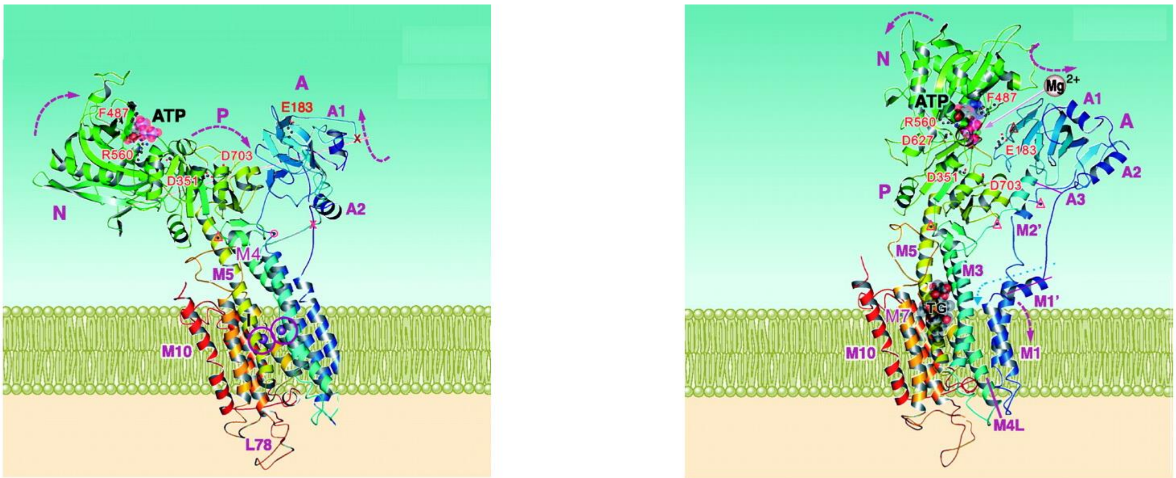

Fig. 1.1

(a) Explain why Ca^{2+} cannot freely cross the plasma membrane. [2]

1. (Hydrophobic) non-polar fatty acids make up the hydrophobic core of the plasma membrane, therefore the membrane is impermeable to Ca^{2+};;
2. Ca^{2+} are charged and are repelled by the hydrophobic core of the plasma membrane, thus Ca^{2+} cannot cross the plasma membrane freely;;

(b) Describe how low intracellular concentrations of Ca^{2+} are maintained by PMCA. [3]

1. Ca^{2+} recognises and binds to the specific binding site of PMCA/the carrier protein;; (extra pt)
2. PMCA undergoes a conformational change and releases Ca^{2+} to the extracellular/other side of the membrane;;
3. Ca^{2+} is transported across the plasma membrane via active transport;;
4. against a concentration gradient;;
5. ATP is hydrolysed/required;;

Various forms of PMCA are expressed in different cell types, including erythrocytes (red blood cells). Erythrocytes rely on Ca^{2+} dependent signalling during their differentiation from hematopoietic stem cells in the bone marrow. The process of erythropoiesis (production of mature red blood cells) is illustrated in Fig. 1.2.

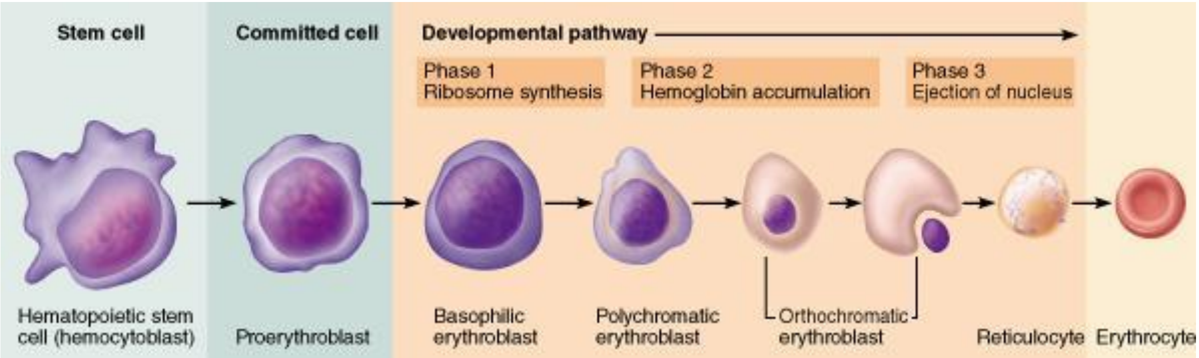

Fig. 1.2

(c) Describe the potency of these hematopoietic stem cells and explain their normal functions. [4]

1. **These hematopoietic/blood stem cells are multipotent;;**
2. **They have the ability to differentiate into a limited range of cell types and so are not pluripotent or totipotent;;**

3. **To maintain and repair the specific tissue (blood) where they are found/where they reside (by replacing worn-out or damaged cells);;**
4. **Can undergo differentiation to form the different types of blood cells – red blood cells and white blood cells (e.g. B lymphocytes, T lymphocytes, natural killer cells, macrophages, platelets etc.);;**
5. **To replace worn-out or damaged red blood cells and white blood cells;;**
6. **Will constantly divide to replace cells such as the red blood cells that are worn out in three to four months;;**

In addition to the loss of the nucleus, cellular organelles such as the endoplasmic reticulum, Golgi apparatus and mitochondria are also lost from the reticulocytes shown in Fig. 1.2.

(d) Outline the structure of the endoplasmic reticulum and Golgi apparatus in typical eukaryotic cells. [2]

1. **The (rough) endoplasmic reticulum <u>consists of a network of sheets</u> called cisternae;;**
   OR
2. **The (smooth) endoplasmic reticulum <u>consists of a network of tubules/tubes</u> called cisternae**
3. **The Golgi apparatus <u>consists of a stack of flattened membrane-bound sacs</u> called cisternae (together with a system of associated vesicles called Golgi vesicles);;**

(e) Since mature mammalian erythrocytes lack mitochondria, suggest how these cells derive energy from glucose. [1]

1. **Mature erythrocytes can still obtain <u>ATP</u> (via substrate level phosphorylation) <u>from glycolysis</u>;;**

[Total: 12]

2 Fig. 2.1 shows two animal cells in different stages of the mitotic cell cycle.

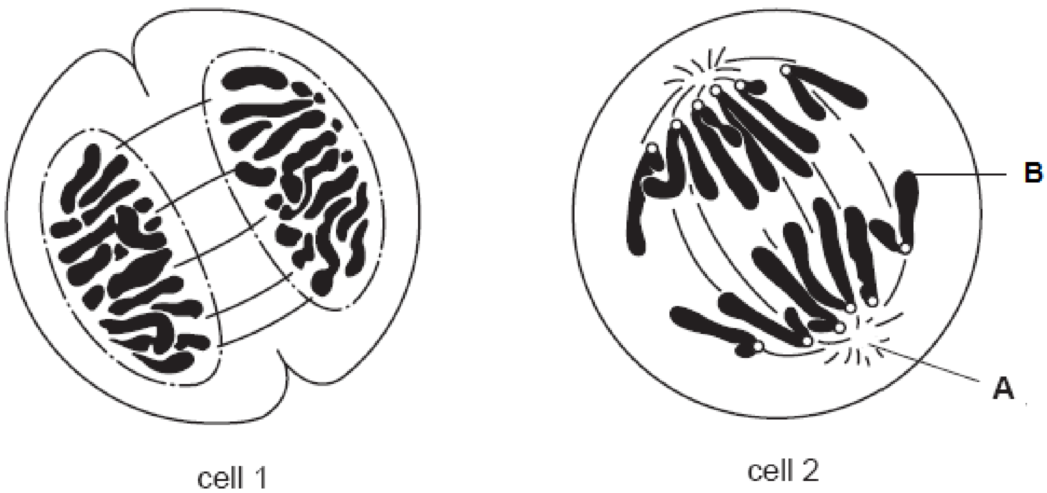

Fig. 2.1

(a) With reference to cell 1,

(i) identify the stage of nuclear division taking place. [1]

1. **Telophase;;**

(ii) describe the events occurring at this stage. [2]

1. **Daughter chromosomes pulled by spindle fibres attached to centromeres reach opposite poles of the cell;;**
2. **Spindle fibres disintegrate;;**
3. **Nuclear envelope reforms around the chromatin (accept: chromosomes) in each daughter cell;;**
   (any 2)

(b) A pair of rod-like structures can be found in region A. Outline the roles of these structures during mitosis in animal cells. [2]

1. **(The two pairs of) centrioles <u>move to the opposite poles of the cell</u> (during prophase) and <u>determine the polarity of the cell</u>;;**
2. **Centrioles act as the microtubule-organising centre (MTOC) – the centrioles <u>produce spindle fibres</u> at the poles towards the equator of the cell;;**
3. **Centrioles <u>organise the synthesis of spindle fibres</u> which lead to the separation of chromatids during cell division;;**

(c) Region B of the chromatid contains non-coding repetitive nucleotide sequences.

(i) Account for the progressive shortening of these sequences after repeated rounds of DNA replication. [2]

1. **DNA polymerase can only add DNA nucleotides to the free 3' (-OH) end of an existing strand;;**
2. **With the removal of the RNA primer from the 5' end of the newly synthesised DNA strand, there is no free 3' (-OH) end for DNA polymerase to add (free) nucleotides to,;;**
   OR
3. **With the removal of the RNA primer from the 5' end of the newly synthesised DNA strand, the RNA primer at the 5' end of the (newly synthesised) DNA strand is removed but not replaced;;**
4. **This results in a daughter strand that is shorter than the parental/template DNA strand.**

(ii) Describe two functions of these non-coding DNA in eukaryotes. [2]

1. **Telomeres ensure genes are not lost or eroded due to the end replication problem with each round of DNA replication, preventing loss of important genetic information;;**
2. **Telomeres protect and stabilise the terminal ends of chromosomes by forming a loop which confers stability to linear chromosomes by preventing accidental fusion of the single-stranded end of one chromosome to the single-stranded end of another chromosome via complementary base pairing;;**
   OR
3. **Telomeres protect and stabilise the terminal ends of chromosomes. Telomeric DNA is bound by specific telomere-specific binding proteins which protect the chromosomal ends from degradation by exonucleases;;**
4. **Telomeres that are critically short trigger apoptosis (programmed cell death);;**
5. **Telomeres allow their own extension, by providing an attachment point for the correct positioning of telomerase;;**
   (any 2)

(d) In 2006, Yamanaka and his colleagues demonstrated in an experiment with mice that induced pluripotent stem (iPS) cells could be produced by genetically reprogramming fully differentiated adult cells. There has been evidence to suggest that these iPS cells exhibit high telomerase reverse transcriptase (TERT) activity and are capable of dividing indefinitely.

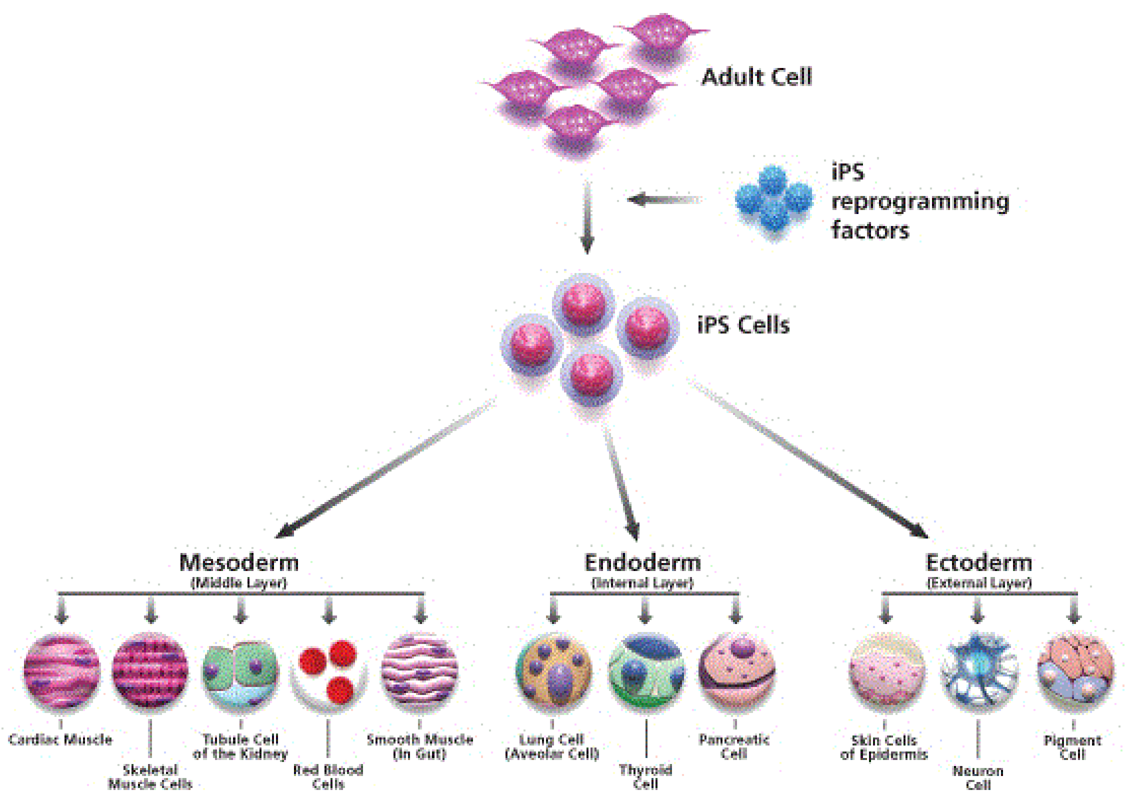

Fig. 2.2

(i) Discuss the role of TERT in enabling the iPS cells to divide indefinitely. [2]

1. **TERT catalyses the <u>elongation of telomeres / maintain length of telomeres / maintain number of telomere repeat sequences</u> by adding telomere repeat sequences to the 3' end of the DNA strand/telomeres;;**
2. **Therefore, the cells do not enter replicative senescence / apoptosis is not triggered;;**

(ii) Suggest an advantage of using iPS cells in research and medical applications. [1]

1. **Since iPS cells are derived from adult cells, the use of iPS cells overcomes ethical issues pertaining to the destruction of human embryos as a source of (pluripotent) stem cells;;**
   OR
2. **Since iPS cells can be derived from adult cells of the patient, the use of iPS cells presents no/low risk of immune rejection;;**

[Total: 12]

3 Antibiotic resistance is rising to dangerously high levels in all parts of the world. A growing list of infections – such as pneumonia, tuberculosis, blood poisoning and gonorrhoea – are becoming harder, and sometimes impossible, to treat as antibiotics become less effective.

Some bacteria are naturally resistant to certain types of antibiotics. However, bacteria may also become resistant either by a genetic mutation or by acquiring resistance from another bacterium.

(a) Outline the process of how a bacterium is able to acquire resistance from another bacterium. [3]

1. **bacterial conjugation;;**
2. **F^+ cell/donor bacterial cell with F factor produces sex pilus to attach itself to F^- cell/recipient cell;;**
3. **A temporary cytoplasmic mating bridge is formed between the two bacterial cells which allows F^+ cell to transfer its F plasmid containing the antibiotic resistance gene to the F^- cell (by rolling circle mechanism);;**

Or

4. **bacterial transformation;;**
5. **A bacterium takes up foreign DNA containing antibiotic resistance gene;;**
6. **The foreign DNA is incorporated into bacterium's own DNA via homologous recombination/through crossing over with a homologous region found on the bacterial chromosome;;**

More than 2 million Americans each year are infected by antibiotic-resistant bacteria, and at least 23,000 die annually from those infections. Antibiotic-resistant bacteria have become a global health crisis and alternative treatments such as Phage Therapy are being considered for combating bacterial infections.

Phage Therapy involves the targeted application of bacteriophages that, upon encounter with specific pathogenic bacteria, can infect and kill them. Phages are currently being used therapeutically to treat bacterial infections that do not respond to conventional antibiotics.

Fig. 3.1 is an electron micrograph showing a phage infecting a bacterium during Phage Therapy.

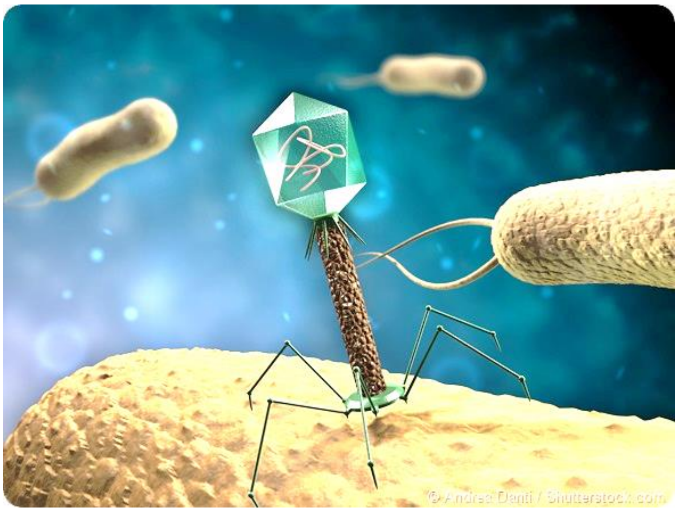

Fig. 3.1

(b) Suggest the reproductive cycle of a phage used for Phage Therapy. [1]

- **lytic cycle**

(c) Describe how a structural feature of the phage allows for targeted application to specific pathogenic bacteria. [2]

1. **attachment sites on its tail fibres;;**
2. **<u>complementary in shape to specific receptor sites</u> on the specific host bacterial cell wall, <u>recognise and adsorb</u> to specific receptor sites on the specific host bacterial cell wall;;** ***(mark for 'specific receptor sites on the specific host bacterial cell wall' once)***

(d) Explain how the use of phages can prevent the spread of bacterial infection. [2]

1. **enzymes coded by the genome of the phage shuts down the bacterium's macromolecular (i.e. DNA, RNA and protein) synthesis;;**
   OR
2. **phage nucleases hydrolyse the bacterial chromosome;;**
   OR
3. **lysozyme breaks down peptidoglycan cell wall;;**
4. **new bacteria cells cannot be synthesized;;**
5. **lysis (death) of host cell occurs upon the release of new phage particles;;**

(e) Suggest characteristics of phages that make them attractive therapeutic agents. [2]

1. **highly specific / more specific than antibiotic;;**
2. **very effective in lysing targeted pathogenic bacteria;;**
3. **typically harmless;;**
4. **will not develop resistance;;**
5. **rapidly modifiable to combat the emergence of newly arising bacterial threats;;**

[Total: 10]

4 Sickle cell anaemia is a recessive genetic disease caused by a mutation that commonly occurs in the DNA, resulting in hydrophobic valine replacing hydrophilic glutamic acid at the 6th amino acid position of the β chain.

(a) State the type of mutation that commonly occurs to result in sickle cell anaemia. [1]

1. (single) base-pair substitution;;

(b) Describe the effects of this change in amino acids on the red blood cells of an individual with the disease. [4]

1. **This results in a change in the tertiary structure/3D conformation of haemoglobin to produce haemoglobin S (HbS) instead of HbA;;**
2. **This decreases the solubility of deoxygenated HbS and at low oxygen concentration, hydrophobic areas of different HbS would stick together;;**
3. **HbS molecules will polymerise and precipitate out of solution to form rigid fibres;;**
4. **the change from HbA to HbS would result in changes to the shape of the red blood cells from circular biconcave shape to become sickle shape;;**

To detect if individuals are afflicted with sickle cell anaemia, restriction fragment length polymorphism (RFLP) analysis can be carried out using gel electrophoresis and Southern Blotting. Restriction enzymes are used to digest the DNA before RFLP analysis and the mutation removes a recognition site of the restriction enzyme *MstII,* as shown in Fig. 4.1. The enzyme's recognition sites on the normal allele and the mutant allele are shown by arrows.

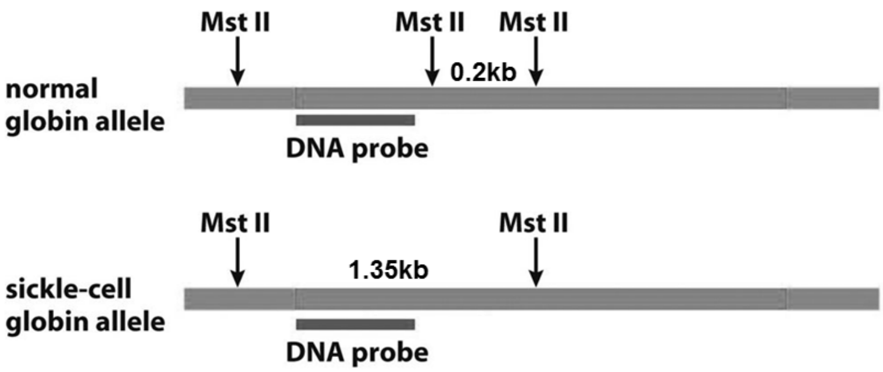

Fig. 4.1

(c) Draw, on Fig. 4.2, the expected band patterns produced by DNA from individuals with sickle cell anaemia. [1]

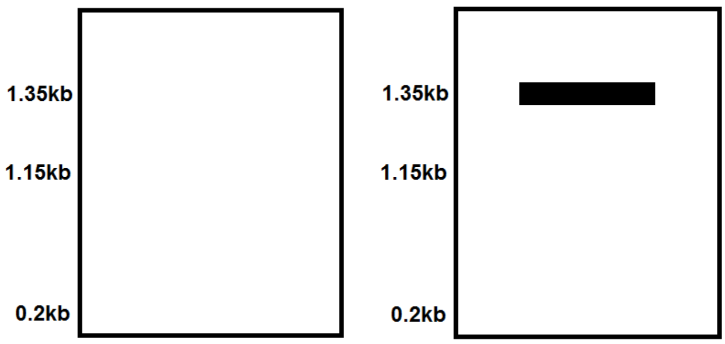

Fig. 4.2

(d) Suggest why it is necessary to carry out Southern Blotting after gel electrophoresis. [2]

1. **After gel electrophoresis is carried out to separate DNA fragments according to size, there might be <u>too many bands to be distinguished individually</u>;;**
2. **Hence, Southern blot is usually carried out to <u>identify the DNA fragment of interest</u>;;**

(e) Outline the role of the DNA probe in Southern Blotting. [1]

1. **The probe is single-stranded to hybridise/complementary base pair with DNA fragments/specific nucleotide sequences on the nitrocellulose paper;;**
2. **The probe is radioactively-labelled so that the DNA fragments will show up as bands after autoradiography/on the photographic X-ray film;;**

[Total: 9]

5 Fig. 5.1 shows awned and awnless rice strains. A long awn is one of the distinct morphological features of wild rice species. It is a long needle-like appendage that is thought to aid in seed dispersal and prevent predation by animals.

The genes *DROOPING LEAF (DL)* and *OsETTIN2 (OsETT2)* are involved in awn formation. Genetic analysis experiments indicate that *DL* and *OsETT2* act independently in awn formation.

Fig. 5.1

A cross between pure-breeding awned and awnless strains produced awned plants in F1.

The F1 plants were then self-pollinated.

In the F2 generation, 658 awned plants and 48 awnless plants were produced.

This control of awn development is an example of epistasis resulting in a ratio that is close to 15:1.

(a) Define the term *locus*. [2]

1. **the position of a gene;;**
2. **on a chromosome or within a DNA molecule;;**

(b) Explain the term epistasis in this context. [3]

1. **The awn formation is controlled by two genes occupying different loci;;**
2. **when either gene has a dominant allele at the gene loci;; (it hides the effect of the other gene)**
   OR
3. **a copy of dominant allele at either gene results in awn development;;**
4. **homozygous for the recessive allele at both genes results in awnless plants/character;;**
   (Such gene interaction is also known as duplicate dominant epistasis)

(c) Use the symbols A, a and B, b to draw a genetic diagram to explain the results shown in the F2 generation. [4]

F1 phenotypes:	**awned plants**	**X**	**awned plants**		
F1 genotypes:	**AaBb**	**X**	**AaBb**		**;;**
Gametes	**AB** **Ab** **aB** **ab**		**AB** **Ab** **aB** **ab**		**;;**

Fertilization

	AB	**Ab**	**aB**	**ab**
AB	**AABB**	**AABb**	**AaBB**	**AaBb**
Ab	**AABb**	**AAbb**	**AaBb**	**Aabb**
aB	**AaBB**	**AaBb**	**aaBB**	**aaBb**
ab	**AaBb**	**Aabb**	**aaBb**	**aabb**

F2 genotype :	**AABB** **AABb** **AaBB** **AaBb** **Aabb** **AAbb** **aaBB** **aaBb**	**:**	**aabb**	**;;**
F2 phenotype :	**awned plants**	**:**	**awnless plants**	**;;**
F2 phenotypic ratio :	**15**	**:**	**1**	

A chi-squared test was carried out on the results of the second cross.

Table 5.1

phenotype	observed number (O)	expected number (E)	$\frac{(O-E)^2}{E}$	
awn	658	**661.9**	**0.0229793**	;;
awnless	48	**44.1**	**0.339381**	;;
total number	706	706	x^2 = **0.3623603**	;;

(d) Complete the five missing values in Table 5.1. [3]

(e) Table 5.2 shows part of the table of probabilities for the chi-squared test.

Table 5.2

degrees of freedom	probability								
	0.995	0.975	0.9	0.5	0.1	0.05	0.025	0.01	0.005
1	.000	.000	0.016	0.455	2.706	3.841	5.024	6.635	7.879

Use Table 5.2 and your calculated value for the chi-squared test to find the probability that the observed ratio of phenotype does not deviate significantly from the expected ratio. [1]

- **Value of p is between 0.5 - 0.9**

(f) State what conclusions may be drawn from the probability found in (e). [2]

(Value of p is between 0.5 - 0.9, more than p = 0.05)

1. **Do not reject the null hypothesis, there is no significant difference between the observed and the expected ratio;;**
2. **The observed ratio of phenotype does not deviate significantly from the expected ratio, any deviation from the expected is due to chance;;**

[Total: 15]

6 Fig. 6.1 shows some stages in mammalian respiration.

glucose

$\downarrow$ **A**

hexose phosphate

$\downarrow$ **B**

triose phosphate

$\downarrow$ **C**

pyruvate

$\downarrow$ **D**

carbon dioxide and water

Fig. 6.1

(a) Name the processes taking place during Stage D and state precisely where they occur. [3]

1. **Link reaction – mitochondrial matrix;;**
2. **Krebs cycle – mitochondrial matrix;;**
3. **Oxidative phosphorylation – inner mitochondrial membrane;;**

(b) Intermediates produced at the end of Stages B and C are important in the conversion of carbohydrates to lipids such as triglycerides. Some of the triose phosphate can be converted into glycerol-3-phosphate, while pyruvate can undergo further reactions to form intermediates required for the synthesis of fatty acids.

(i) Describe the formation of triglycerides. [3]

1. **A triglyceride is formed by condensation reactions between 1 glycerol and 3 fatty acids;;**
2. **Each of glycerol's hydroxyl/–OH groups condenses with the carboxyl/ –COOH group of a fatty acid;;**
3. **In each condensation reaction, one water molecule is removed, resulting in the formation of an ester bond/linkage;;**

(ii) State two roles of triglycerides in living organisms. [2]

1. **Triglycerides serve as a good energy source;;**
2. **Triglycerides are a weight efficient means for organisms to store energy / serve as good energy storage molecules;;**
3. **Triglycerides serve as a good source of metabolic water;;**
4. **Triglycerides are good thermal insulators that reduce / prevent excessive heat loss from the body;;**
5. **Provide buoyancy to aquatic animals;;**
   (any 2)

(c) The first reaction in Stage A is catalysed by the enzyme hexokinase. It has been observed that hexokinase is bound to the outer mitochondrial membrane in muscle cells which undergo high rates of glycolysis.

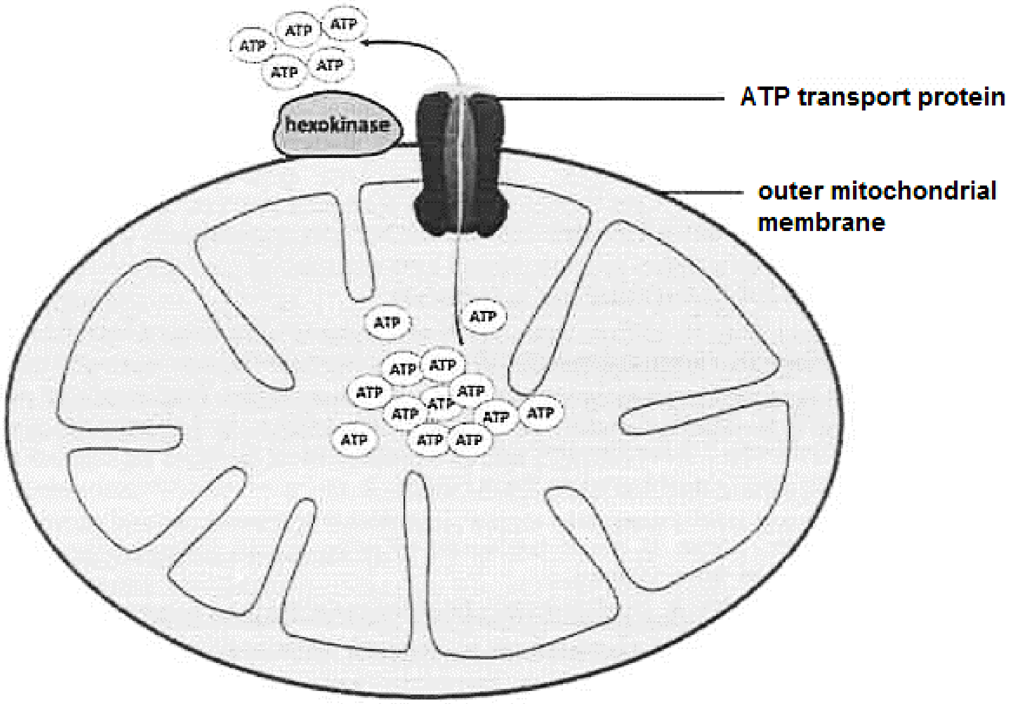

Fig. 6.2

With reference to the role of mitochondria and Fig. 6.2, suggest how the association of hexokinase with mitochondria can lead to high rates of glycolysis. [2]

1. **Mitochondria are the site of aerobic respiration to synthesise ATP;;**
2. **Due to the <u>close proximity</u> of hexokinase to the mitochondria, <u>ATP produced</u> by the mitochondria <u>can easily be used</u> by hexokinase <u>to phosphorylate glucose</u>;; increasing the rate of glycolysis.**

Fig. 6.3 shows an electron micrograph of a mitochondrion.

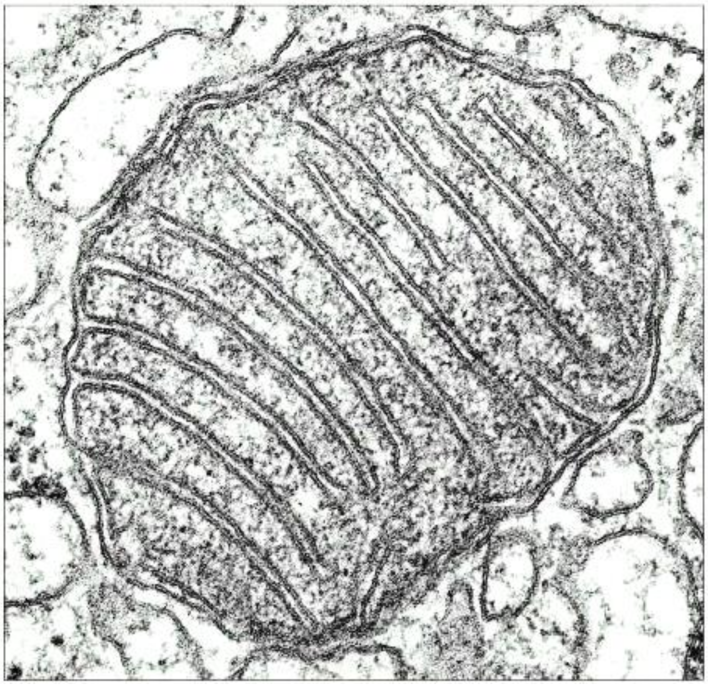

Fig. 6.3

(d) With reference to features visible in Fig. 6.3, outline how the structure of the mitochondrion is adapted for its function. [2]

1. **The inner mitochondrial membrane is <u>highly folded</u>, providing <u>a large surface area</u> where stalked particles, enzymes and electron carriers of the electron transport chain (ETC) *(any 1 e.g.)* needed for <u>aerobic respiration</u> can be located;;**
2. **The mitochondrion is enclosed by <u>double membranes</u> separated by (an extremely narrow fluid-filled space) intermembrane <u>space</u>, allowing for <u>compartmentalisation</u> within the mitochondrion / specialised metabolic pathways to take place in different areas;;**

[Total: 12]

7 Fig. 7.1 shows a flower of *Lilium polyphyllum*, a lily that grows in the Himalayan mountains. This species is cross-pollinated by insects.

Fig. 7.1

(a) Plants of this species that grow at low altitudes produce flowers 60 days before the plants of the same species that grow at high altitudes. Scientists think that plants of *L. polyphyllum* growing at high altitudes may evolve into a new species.

Explain how natural selection could lead to the evolution of a new species of lily. [5]

1. **The subpopulations / low and high altitudes of *L. polyphyllum* become physiologically isolated leading to sympatric speciation;;**
2. **The subpopulations did not interbreed (reproductive isolation due to differing flowering times) and thus gene flow was disrupted (resulting in different species);;**
3. **The subpopulations were exposed to different environments and were thus subjected to different selection pressures;;**
4. **Since there was variation within the subpopulations due to spontaneous mutation,;;**
5. **individuals with favourable characteristics were at a selective advantage and can survive to maturity, (undergo fertilisation / mate), reproduce and passed on their favourable alleles / genes (R:traits) to their offspring (or vice versa);;**
6. **Over successive generations, evolutionary changes occurred independently in each subpopulation;;**
7. **New species of lily have thus arisen by descent with modifications from ancestral species by accumulation of modifications as the population of lily adapt to the new environment;;**

(b) In order for natural selection to occur a population must show phenotypic variation. Explain why variation is important in natural selection. [2]

1. **Variation describes the <u>differences in characteristics</u> shown by individuals belonging to the same species due to presence of different alleles in the individuals;;**
2. **Variation is the <u>raw material</u>, <u>presence of different alleles</u> leading to difference in characteristics, for natural selection to act on;;**
3. **Resulting in <u>differential reproductive success</u>;;**

(c) Fungi were often classified as different species according to their visible reproductive structures. *Penicillium dodgei* and *Eupenicillium brefeldianum* were classified as different species because they had different types of spores.

However, recently it was recognised that the spores of *P. dodgei* were asexual spores, while those of *E. brefeldianum* were sexual spores. A comparison of the DNA of these two fungi shows that they are the same species. This fungus is now known as *Penicillium brefeldianum*.

Outline how DNA analysis can show that *P. dodgei* and *E. brefeldianum* are the same species. [2]

1. **(Homologous DNA sequences) have few differences in DNA bases / sequences;;**
2. **Homologous DNA sequences are identical / more similar in length;;**
3. **Genes are same;;**

(d) Describe the advantages of using DNA analysis in determining homology between *P. dodgei* and *E. brefeldianum*. [3]

1. **<u>Unambiguous</u> and <u>objective</u>. A, T, G, C are <u>easily recognized / one cannot be confused with another</u>. They are <u>not dependent on subjective judgements / observations involving qualitative differences</u>;;**
2. **<u>Quantifiable</u> and <u>can be converted to numerical form and open to statistical and analysis</u>;;**
3. **<u>Homologous regions of DNA from different species provides many points of comparison</u> as each nucleotide position is a point of comparison;; (Each nucleotide position along a stretch of DNA represents an inherited character in the form of one of four DNA bases.)**

[Total: 12]

8 Fig. 8.1 shows part of the immune response to the first infection by a bacterial pathogen that has entered the body through the lining of a bronchiole. J and K are stages in the immune response.

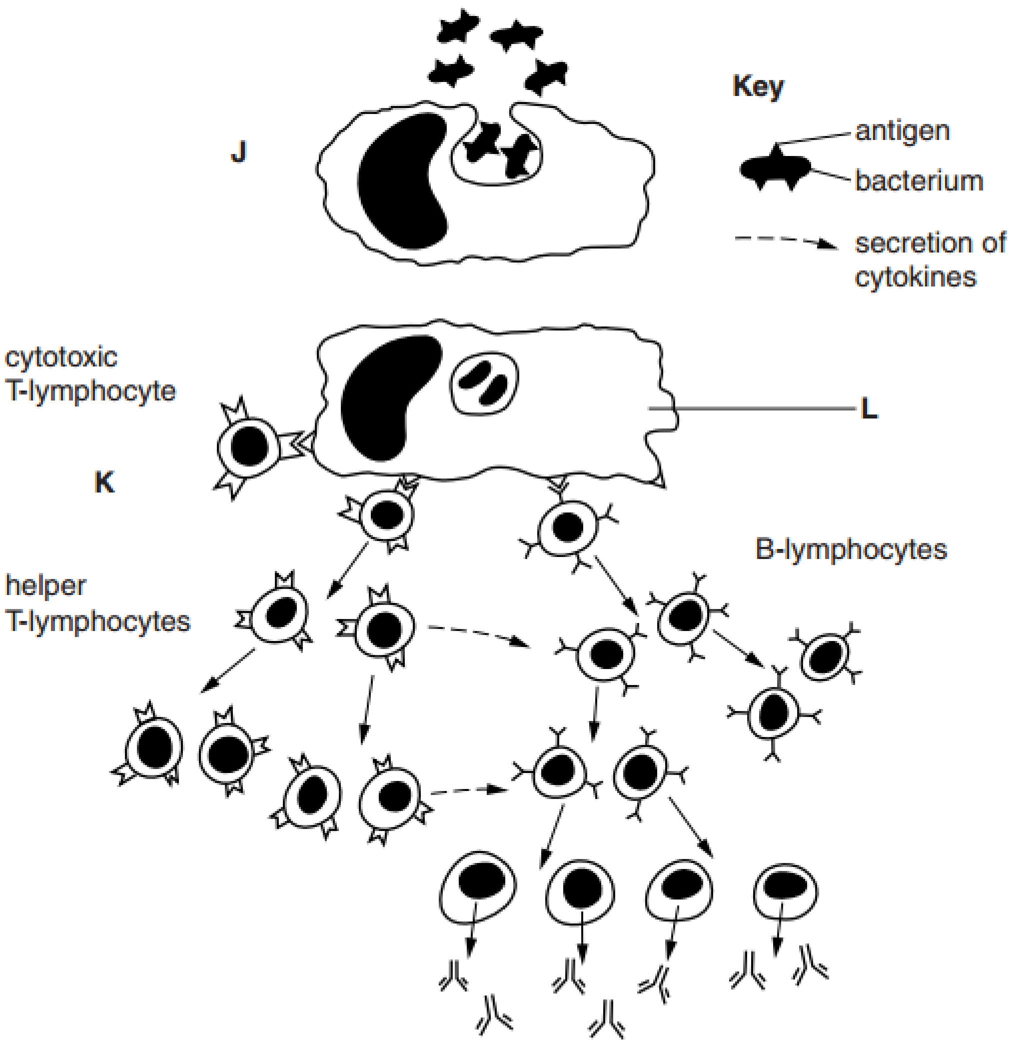

Fig. 8.1

(a) (i) State the process happening at stage J. [1]

1. Phagocytosis / endocytosis;;

(ii) Explain the role of cell L at stage K in the immune response. [2]

1. Digestion of bacteria to destroy bacteria;;
2. Antigen presentation on cell surface;;
3. Clonal selection / activation of specific B / T-lymphocytes;;

(b) With reference to Fig. 8.1, explain how the response to a second infection by this bacterial pathogen differs from the first. [3]

1. **Memory (B / T) cells <u>quickly divide by mitosis to form large numbers</u> of effector B / T-lymphocytes (e.g. B-lymphocytes and helper / cytotoxic T-lymphocytes) upon re-exposure to the same antigen;;**
   OR
2. **There is an <u>increased in number</u> of helper / cytotoxic T-lymphocytes specific for this pathogen;;**
3. **There is faster response (for second and subsequent contacts);;**
4. **which increases chances of coming across more pathogens / APCs;;**
5. **<u>Faster production</u> of B-lymphocytes / plasma cells / antibodies / helper T-lymphocytes / cytotoxic T-lymphocytes / cytokines;;**
6. **<u>Greater concentration</u> of antibodies or greater numbers of B / plasma cells;;**
7. **Pathogen removed / killed faster;;**
8. **Person does not become ill / no symptoms;;**

B-lymphocytes have antibodies located on their external surface. When B-lymphocytes become plasma cells they then secrete antibodies.

Fig. 8.2 shows how the enzyme papain digests an antibody to obtain three fragments.

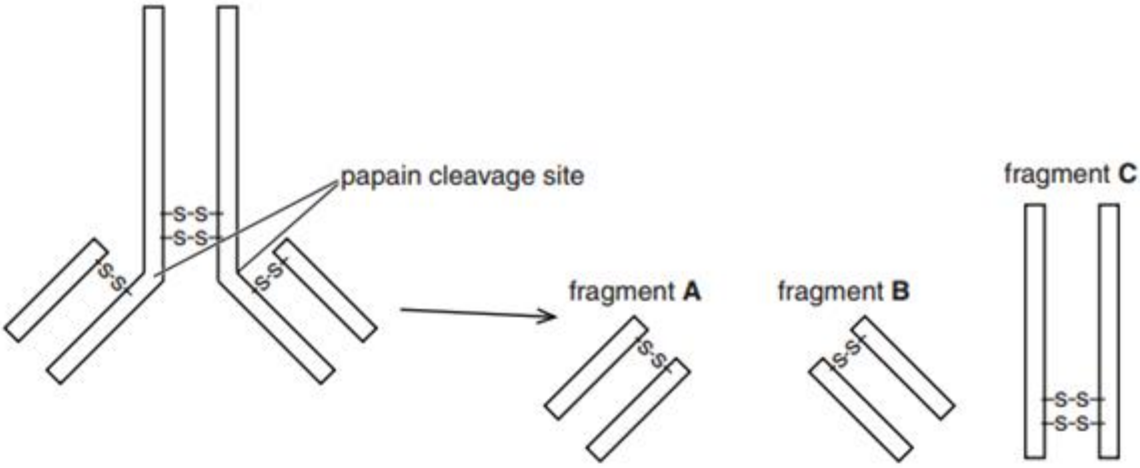

Fig. 8.2

(c) The three fragments, A, B and C still retain their ability to function.

State the function of:

(i) fragments A and B. [1]

1. **Antigen binding sites / bind to antigen / both bind to same (type of) antigen;;**

(ii) fragment C. [1]

1. **Binding to phagocyte / monocyte / macrophage / neutrophil / B-lymphocyte / named cell type with Fc receptor;;**

(d) There are various ways in which the effectiveness of immune responses can be reduced.

Suggest how each of the following reduces the effectiveness of an immune response.

(i) Some pathogens are covered in cell surface membranes from their host. [1]

1. **Pathogens not recognised as non-self / foreign (or vice versa);;**

(ii) B-lymphocytes do not mature properly and do not recognise any antigens. [1]

1. **No antibodies / plasma cells / memory B cells produced;;**
2. **No humoral response;;**
3. **No antigen presentation by B-lymphocytes;;**

[Total: 10]

9 Reef-building corals are marine invertebrates closely related to jellyfishes and are found in shallow, clear tropical seas. The corals secrete an exoskeleton of calcium carbonate that becomes the underlying structure of the coral reef.

Zooxanthellae are a group of unicellular photosynthetic algae that live inside the cells of reef-building corals. The relationship is beneficial to both the zooxanthellae and the coral.

(a) Evidence shows that the relationship between zooxanthellae and reef-building corals has evolved by free-living algae invading corals that did not contain algae. [1]

(i) Corals that do not need zooxanthellae can live at a greater depth that reef-building corals.
Explain why. [3]

1. **Corals without zooxanthellae do not need to rely on light;;**
2. **The corals may have different feeding methods;;**
3. **Reef-building corals with zooxanthellae need light for them to photosynthesise;;**
4. **As depth increases, less light penetration/more light absorbed by the water;;**

(ii) Suggest how the zooxanthellae may benefit in two ways from their association with the corals. [2]

1. **The corals provide zooxanthellae with carbon dioxide for photosynthesis;;**
2. **Protection from predation;;**
3. **Protection from extreme conditions;;**
4. **The corals provide a physical support for zooxanthellae to absorb light;;**
5. **Nitrogen from the coral's nitrogenous waste supports algal growth;;**

Under conditions of environmental stress, the relationship between the reef-building corals and zooxanthellae can break down. Loss of zooxanthellae and the subsequent whitening that occurs, as shown in Fig. 9.1, is known as coral bleaching. Coral bleaching can lead to the death of the coral.

Fig. 9.1

(b) State one reason why permanent loss of zooxanthellae can lead to death of the coral. [1]

1. **Loss of major food source/decreased source of food;;**
2. **Less organic compounds (e.g. sugars);;**
3. **Loss of protective algal layer for corals from harmful effects of sunlight;;**

(c) One type of environmental stress that can cause coral bleaching is an increase in sea temperature.

Suggest why areas of sea with reef-building corals are particularly susceptible to increased temperature as a result of global climate change. [2]

1. **Coral reefs grow in shallow/tropical seas;;**
2. **Shallow water heats up more rapidly/surface waters are warmer than deeper water;;**
3. **Temperature increases may be greater near the equator;;**

[Total: 8]

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